SUMMARY · MK 5 DECISION MAKING & NEGOTIATION · WEEK 7

Analytic Hierarchy Process (AHP)

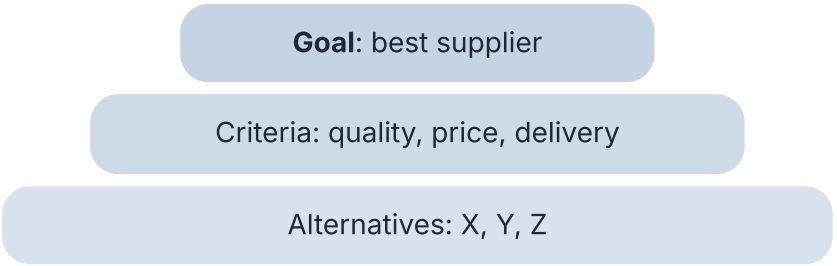
Saaty's AHP: hierarchy of goal–criteria–alternatives, pairwise comparisons on a 1–9 scale, weights via normalisation/eigenvector, consistency index and ratio (CR ≤ 0.10), global synthesis, comparison with SMART, and criticisms (rank reversal).

Material

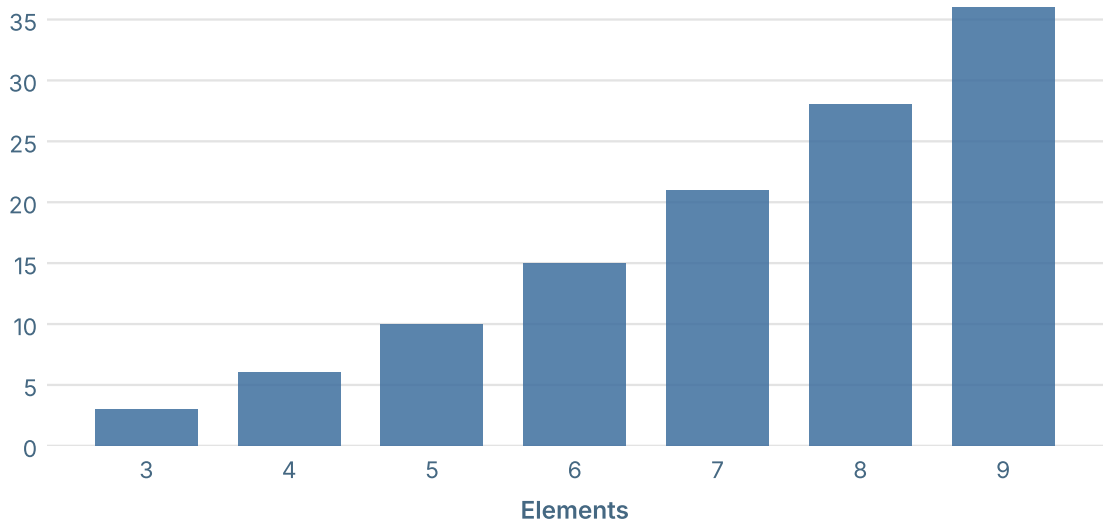
1 Hierarchy and Saaty's scale

Steps: hierarchy, pairwise criteria matrix, weights and consistency, pairwise alternatives per criterion, global synthesis, sensitivity. Scale 1 (equal) to 9 (extreme); reciprocal matrix; n(n−1)/2 judgements.

2 The supplier hierarchy Deep dive



Pairwise comparisons n(n−1)/2



3 Computing weights and consistency

Example criteria quality, price, delivery with judgements 3, 5, 3: weights 0.633 / 0.260 / 0.106; λmax ≈ 3.039, CI 0.019, CR 0.033 (≤ 0.10 is acceptable; RI n=3 0.58, n=4 0.90, n=5 1.12). Synthesis: supplier X 0.469, Y 0.288, Z 0.242.

4 AHP calculator and global synthesis Deep dive

Global supplier priorities



Totals: X 0.469 · Y 0.288 · Z 0.242.

X wins on quality, the heaviest criterion.

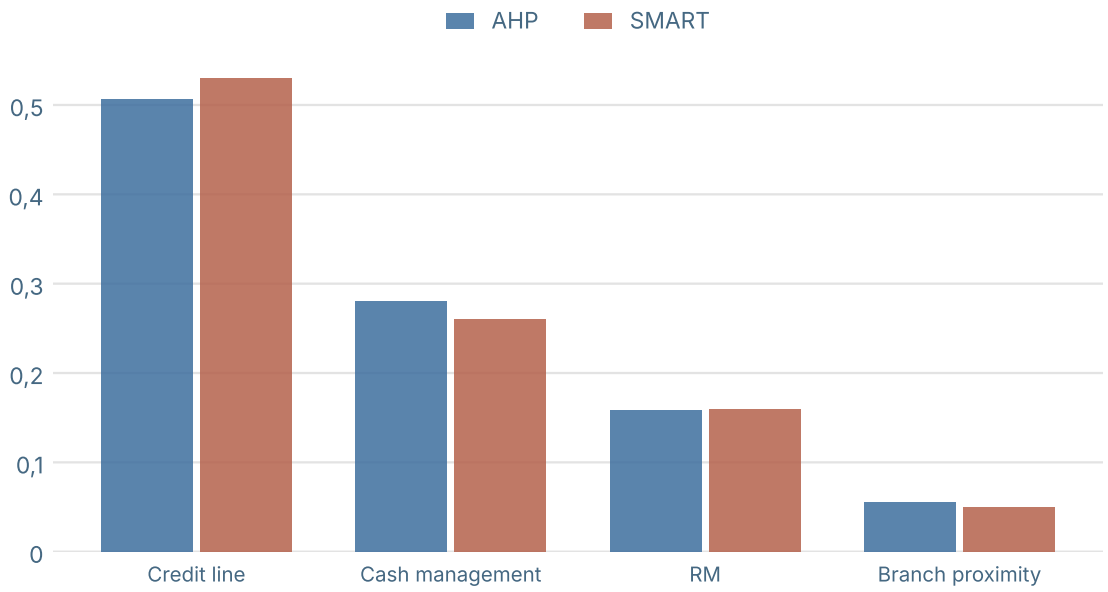
5 AHP vs SMART, and criticisms

SMART: direct ratings and swing weights — fast and transparent. AHP: pairwise judgements with a consistency check, good for qualitative criteria and groups (geometric mean), but many comparisons, scale issues and rank reversal.

6 Answer: the BUMDes exercise with AHP

Deep dive

BUMDes weights: AHP vs SMART



$\lambda_{max} \approx 4.008$, CI 0.003, CR ≈ 0.003

AHP weights nearly match SMART's, so any difference comes from how alternatives are scored.

Key takeaways

1. Goal → criteria → alternatives.
2. 1–9 scale, reciprocal.
3. Normalise columns, average rows.
4. CR ≤ 0.10.
5. Global = Σ weight × local.
6. Beware rank reversal.

Case studies

Case & worked answers

Exercise: the BUMDes case with AHP

Redo the BUMDes bank choice with AHP (example judgements given), check CR with RI 0.90, compare banks pairwise per criterion, and compare the result with SMART — methods shape outcomes, so sensitivity is essential.

Worked examples

Problem 1 **Medium** **Bobot AHP (matriks konsisten)**

Kriteria A, B, C. A vs B = 2, A vs C = 4, B vs C = 2. Hitung bobot dengan normalisasi kolom.

1. Matriks: baris A (1, 2, 4); B (1/2, 1, 2); C (1/4, 1/2, 1).
2. Jumlah kolom: 1,75; 3,5; 7.
3. Normalisasi baris A: 1/1,75 = 0,571; 2/3,5 = 0,571; 4/7 = 0,571 → rata-rata 0,571.
4. Dengan cara sama: B = 0,286; C = 0,143.
5. Semua kolom identik → sangat konsisten (CR = 0), karena 2 × 2 = 4 (A/B × B/C = A/C).

Answer: A 0,571; B 0,286; C 0,143; CR = 0.

Problem 2 Challenging **Mendeteksi penilaian tidak konsisten**

Penilai menulis: A vs B = 3, B vs C = 3, tetapi C vs A = 2 (C lebih penting dari A). Apakah konsisten? (RI n = 3 adalah 0,58.)

1. Logikanya: A > B dan B > C, maka seharusnya A > C (sekitar 9). Tetapi penilai bilang C > A — bertentangan.
2. Jika dihitung, bobot menjadi hampir rata (A 0,37; B 0,33; C 0,30) dan λmax ≈ 4,01.
3. CI = (4,01 − 3) ÷ 2 = 0,50; CR = 0,50 ÷ 0,58 = 0,87.

Answer: CR ≈ 0,87 jauh di atas 0,10 → tidak konsisten. Minta penilai meninjau ulang perbandingan A vs C.

Practice questions

Try answering first. The answer key is below each question.

1. If A vs B = 5, B vs A = ...

A. 5
B. −5
C. 1/5
D. 0

Answer: C. Reciprocal matrix.

2. Acceptable CR is...

A. ≤ 0.01
B. ≤ 0.10
C. ≤ 0.50
D. ≥ 1

Answer: B. Saaty.

3. For 6 criteria, comparisons needed…

A. 6
B. 12
C. 15
D. 36

Answer: C. 6×5/2.

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4. Rank reversal means…

A. Inverted weights
B. Adding an irrelevant option can change the ranking
C. Negative CR
D. Non-reciprocal

Answer: B. A key criticism.

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5. RI for n = 4 is…

A. 0.58
B. 0.90
C. 1.12
D. 1.24

Answer: B. Used in the BUMDes exercise.

Sources

- [for Students] 07 Decision Analysis: AHP (MK5) — slide berupa gambar; contoh disusun ulang
- Saaty, The Analytic Hierarchy Process
- Goodwin & Wright, Bab 16 (AHP)